Inverse Functions

Undo the function — find it, prove it, graph it, and get its domain & range

The big idea

A function f takes an input x to an output y. Its **inverse** f^{-1} goes back: it takes y to x.

$$x \xrightarrow{f} y \qquad y \xrightarrow{f^{-1}} x \qquad \text{so} \qquad f(a) = b \iff f^{-1}(b) = a.$$

Every point (a, b) on f becomes the point (b, a) on f^{-1}: **inputs and outputs trade places.**

Simple case first. "Add 3" is undone by "subtract 3": if $f(x) = x + 3$ then $f^{-1}(x) = x - 3$. Check: $f(4) = 7$ and $f^{-1}(7) = 4$.
"Double, then add 3" is undone by doing the opposite steps in the **opposite order**: subtract 3, then halve. Like putting on socks then shoes — you take off the shoes first.

x	0	1	2	3
$f(x) = 2x + 3$	3	5	7	9

$\Longrightarrow$

x	3	5	7	9
$f^{-1}(x)$	0	1	2	3

Watch out: $f^{-1}(x)$ means the *inverse function*. It is **not** $\dfrac{1}{f(x)}$. The -1 is not an exponent here.

1 How to find the inverse — four steps

1. Replace $f(x)$ with y.
2. **Swap** x and y.
3. **Solve** for y.
4. Replace y with $f^{-1}(x)$ and **state its domain**.

Step 2 is the whole idea: swapping x and y is exactly "inputs and outputs trade places." Steps 3 and 4 just tidy up the answer.

Example 1 · a line

Find the inverse of $f(x) = 2x + 3$.

$y = 2x + 3$ Step 1

$x = 2y + 3$ Step 2: swap

$x - 3 = 2y \;\Rightarrow\; y = \dfrac{x-3}{2}$ Step 3: solve

$f^{-1}(x) = \dfrac{x-3}{2}$, domain: all real numbers Step 4

Simple case check: $f(1) = 5$, and $f^{-1}(5) = \frac{5-3}{2} = 1$. ✓

Example 2 · a rational function

Find the inverse of $f(x) = \dfrac{2x+1}{x-3}$.

Swap: $x = \dfrac{2y+1}{y-3}$. Multiply by $(y-3)$: $xy - 3x = 2y + 1$. Collect the y terms on one side: $xy - 2y = 3x + 1$.

Factor out y: $y(x-2) = 3x + 1$, so

$$f^{-1}(x) = \frac{3x+1}{x-2}, \qquad x \neq 2.$$

Simple case check: $f(4) = \frac{9}{1} = 9$, and $f^{-1}(9) = \frac{28}{7} = 4$. ✓

Example 3 · a square root — the domain matters

Find the inverse of $f(x) = \sqrt{x-2}$.

Swap: $x = \sqrt{y-2}$. Square both sides: $x^2 = y - 2$, so $y = x^2 + 2$.

But x here was a square-root output, and square roots are never negative, so $x \geq 0$:

$$f^{-1}(x) = x^2 + 2, \qquad x \geq 0.$$

Without "$x \geq 0$" you would get a whole parabola, which is *not* the inverse (see the graph in Part 3).

2 How to prove two functions are inverses

The test — you must check BOTH orders

f and g are inverse functions exactly when

$f(g(x)) = x$ for every x in the domain of g and $g(f(x)) = x$ for every x in the domain of f.

In words: doing one and then the other **always brings you back to where you started**.

Why this is the right test. Suppose $f(a) = b$. If g undoes f, then $g(b) = a$, i.e. $g(f(a)) = a$. The same thinking with the roles switched gives $f(g(b)) = b$. So "undoing" means exactly the two equations above.

Proof 1 · $f(x) = 2x + 3$ **and** $g(x) = \dfrac{x-3}{2}$

$$f(g(x)) = 2 \cdot \frac{x-3}{2} + 3 = (x-3) + 3 = x. \qquad g(f(x)) = \frac{(2x+3)-3}{2} = \frac{2x}{2} = x.$$

Both compositions equal x for all real x, so $g = f^{-1}$. ■

Proof 2 · $f(x) = \dfrac{2x+1}{x-3}$ **and** $g(x) = \dfrac{3x+1}{x-2}$

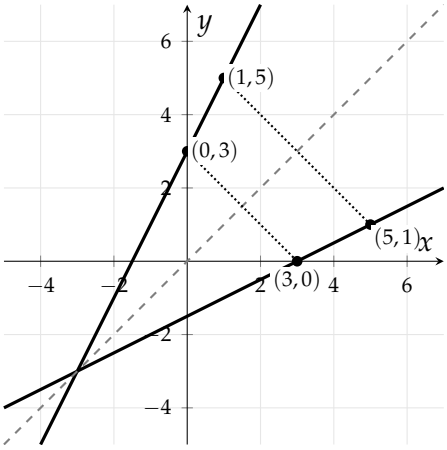
$$f(g(x)) = \frac{2 \cdot \frac{3x+1}{x-2} + 1}{\frac{3x+1}{x-2} - 3} = \frac{\frac{2(3x+1)+(x-2)}{x-2}}{\frac{(3x+1)-3(x-2)}{x-2}} = \frac{6x+2+x-2}{3x+1-3x+6} = \frac{7x}{7} = x \quad (x \neq 2).$$

$$g(f(x)) = \frac{3 \cdot \frac{2x+1}{x-3} + 1}{\frac{2x+1}{x-3} - 2} = \frac{\frac{3(2x+1)+(x-3)}{x-3}}{\frac{(2x+1)-2(x-3)}{x-3}} = \frac{6x+3+x-3}{2x+1-2x+6} = \frac{7x}{7} = x \quad (x \neq 3).$$

Both orders give x, so they are inverses. ■

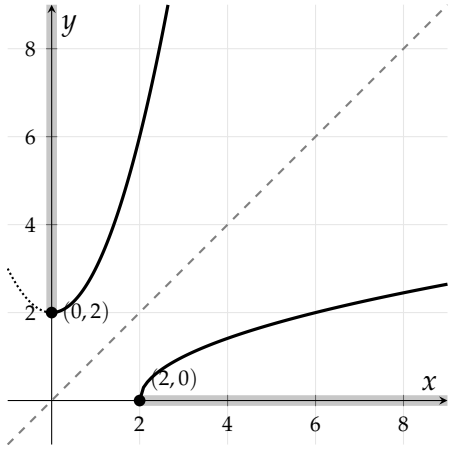
Why one order is not enough. Take $f(x) = x^2$ and $g(x) = \sqrt{x}$. Then $f(g(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$ — looks fine. But $g(f(x)) = \sqrt{x^2} = |x|$, and $g(f(-3)) = 3 \neq -3$. So they are **not** inverses — unless we restrict f to $x \geq 0$ (Part 4).

3 The graph: a mirror image over the line $y = x$



$f(x) = 2x + 3$ $f^{-1}(x) = \frac{x-3}{2}$ $y = x$

Figure 1. Each point (a, b) on f has a partner (b, a) on f^{-1}.



$f(x) = \sqrt{x-2}$ $f^{-1}(x) = x^2 + 2,\ x \geq 0$ $y = x$

Figure 2. Shaded strips: domain of f (blue) and range of f^{-1} (red).

Fold the paper along $y = x$: the graph of f lands exactly on the graph of f^{-1}.

Why? The points (a, b) and (b, a) are mirror images in $y = x$:

- their midpoint $\left(\frac{a+b}{2}, \frac{a+b}{2}\right)$ **lies on** $y = x$;
- the segment joining them has slope $\frac{a-b}{b-a} = -1$, so it is **perpendicular** to $y = x$.

Every point of f is reflected, so the whole graph is reflected. **Slope bonus:** 2 and $\frac{1}{2}$ are reciprocals — steep becomes flat.

Domain and range trade places, too (Figure 2).

$f(x) = \sqrt{x-2}$: domain $[2, \infty)$, range $[0, \infty)$.

$f^{-1}(x) = x^2 + 2,\ x \geq 0$: domain $[0, \infty)$, range $[2, \infty)$.

The dotted left half of the parabola is **not** part of f^{-1} — it is not the mirror image of any point of f. That is why we wrote "$x \geq 0$" in Example 3.

4 Domain and range of the inverse

The rule

$$\text{Domain of } f^{-1} = \text{Range of } f \qquad \text{Range of } f^{-1} = \text{Domain of } f$$

Reason: f^{-1} uses the outputs of f as its inputs, and gives back the inputs of f as its outputs.

How to find them — the safe method

1. Find the domain and range of f **first** (look at the graph, or ask "which x are allowed?" and "which y can come out?").
2. **Swap** them: that gives the domain and range of f^{-1}.
3. Find the formula for f^{-1}, then **attach the domain from step 2** to it.

Don't trust the formula alone. In Example 3 the formula $x^2 + 2$ works for every real x, but the inverse only has domain $[0, \infty)$ — because that is the range of $f(x) = \sqrt{x-2}$. Always get the domain of f^{-1} from the range of f.

Example 4 · rational function

$f(x) = \dfrac{2x+1}{x-3}$: domain $x \neq 3$ (can't divide by 0). Horizontal asymptote $y = \frac{2}{1} = 2$, and $f(x) = 2$ has no solution, so range $y \neq 2$.
Therefore $f^{-1}(x) = \dfrac{3x+1}{x-2}$ has **domain** $x \neq 2$ and **range** $y \neq 3$. The formula agrees: it is undefined at $x = 2$, and its horizontal asymptote is $y = 3$. ✓

Example 5 · when there is no inverse — restrict the domain

$f(x) = x^2$ is **not one-to-one**: $f(2) = f(-2) = 4$, so $f^{-1}(4)$ would have to be both 2 and -2. On the graph, the horizontal line $y = 4$ hits the parabola twice (**horizontal line test** fails).
Fix: keep only half. With domain $x \geq 0$: range $[0, \infty)$, so $f^{-1}(x) = \sqrt{x}$ with domain $[0, \infty)$ and range $[0, \infty)$.
With domain $x \leq 0$ instead: $f^{-1}(x) = -\sqrt{x}$, domain $[0, \infty)$, range $(-\infty, 0]$.

Question	**About f**	**About f^{-1}**
Point	(a, b)	(b, a)
Domain	D	the range of f
Range	R	the domain of f
Graph	—	reflection of f over $y = x$
Exists?	only if f is one-to-one	passes the horizontal line test
Proof	$f(f^{-1}(x)) = x$ **and** $f^{-1}(f(x)) = x$	

5 Practice

Show your work. When you find an inverse, do a simple-case check: pick an easy x, find $f(x)$, and make sure f^{-1} sends it back.

A. Warm-up EASY

1. If $f(3) = 7$, what is $f^{-1}(7)$? If $f^{-1}(-2) = 5$, what is $f(5)$?

2. The table shows f. Find $f^{-1}(11)$ and $f^{-1}(5)$, and make a table for f^{-1}.

x	1	2	3	4
$f(x)$	5	8	11	14

3. Find $f^{-1}(x)$ for $f(x) = 4x - 5$.

4. Find $f^{-1}(x)$ for $f(x) = \dfrac{x+6}{3}$.

B. Prove it (check both orders) MEDIUM

5. Prove that $f(x) = 3x - 7$ and $g(x) = \dfrac{x+7}{3}$ are inverses.

6. Are $f(x) = x^3 + 1$ and $g(x) = \sqrt[3]{x-1}$ inverses? Prove your answer.

7. Are $f(x) = 2x + 4$ and $g(x) = \dfrac{x}{2} - 4$ inverses? Prove your answer.

C. Domain and range of f and f^{-1} MEDIUM HARD

For each function: (a) domain and range of f, (b) the formula for f^{-1}, (c) domain and range of f^{-1}.

8. MEDIUM $f(x) = \sqrt{x+3}$

9. MEDIUM $f(x) = (x-2)^2 + 1, \quad x \geq 2$

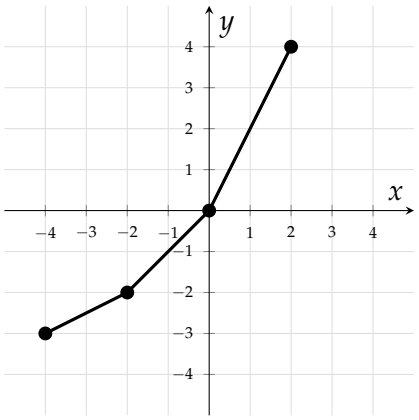
10. MEDIUM $f(x) = \dfrac{x+1}{x-4}$

11. HARD $f(x) = -\sqrt{4-x}$

12. HARD $f(x) = x^2 - 6x$ is not one-to-one. Restrict its domain to the largest interval that contains $x = 5$ on which f *is* one-to-one. Then find f^{-1} with its domain and range.

D. Graphs MEDIUM

13. The graph of f is shown (blue, through $(-4,-3)$, $(-2,-2)$, $(0,0)$, $(2,4)$). Draw $y = x$, then sketch f^{-1} by swapping the coordinates of each marked point. State the domain and range of f and of f^{-1}.



14. (a) The point $(-4, 1)$ is on the graph of f. Which point must be on the graph of f^{-1}?
(b) For $f(x) = 2x - 3$, find the point where the graphs of f and f^{-1} cross.

E. Challenge HARD

15. Show that $f(x) = \dfrac{3x+2}{x-3}$ is its own inverse. What does that tell you about its graph?

16. Explain why $f(x) = x^2$ has no inverse on all real numbers, but $g(x) = x^2$, $x \geq 0$, does. Give the domain and range of g^{-1}.

Before you hand it in: for every inverse you found, did you (1) check one easy number both ways, and (2) write the domain of f^{-1} as the **range of** f?